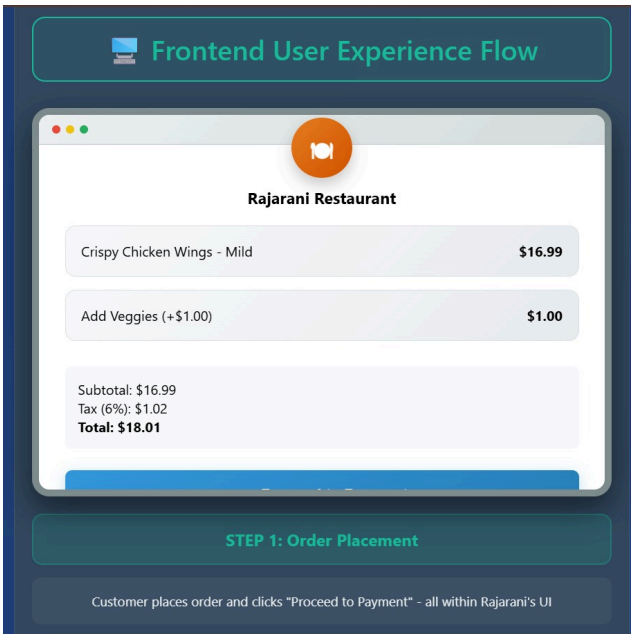
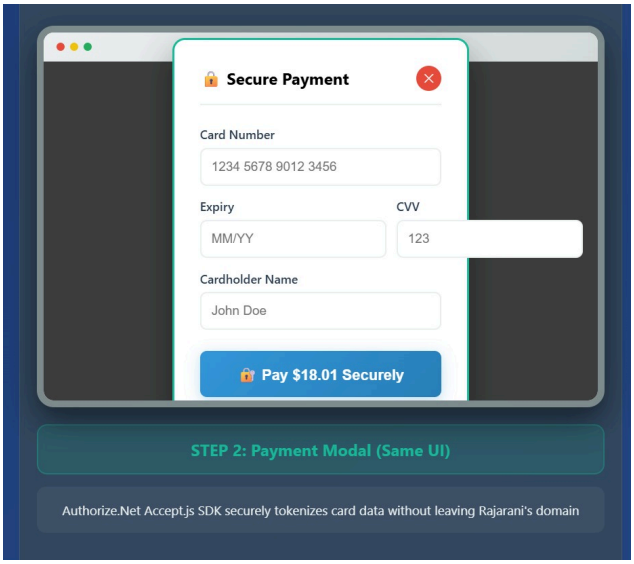
PAYMENT GATEWAY ARCHITECTURE





Backend Processing Architecture (Detailed)

Node.js API Server

POST /api/process-payment

```
const processPayment = async (req, res) => { try { // 1. Validate
incoming token const { token, amount, orderId } = req.body; // 2.
Verify amount matches order const order = await
Order.findById(orderId); if (order.total !== amount) { throw new
Error('Amount mismatch'); } // 3. Process with Authorize.Net const
result = await authorizeNetAPI.charge({ token, amount, orderId });
// 4. Success response res.json({ success: true, transactionId:
result.id }); } catch (error) { // 5. Error handling
res.status(400).json({ success: false, error: error.message }); }
};
```

Authorize.Net API Integration

Payment Gateway Processing:

- → API Login ID & Transaction Key authentication
- → Sandbox/Production environment switching
- → Transaction request formatting
- → Real-time response processing
- → Webhook endpoint for notifications

https://api.authorize.net/xml/v1/request.api

```
const authorizeNetRequest = { createTransactionRequest: {
merchantAuthentication: { name: process.env.AUTH_NET_LOGIN_ID,
transactionKey: process.env.AUTH_NET_TRANS_KEY },
transactionRequest: { transactionType: "authCaptureTransaction",
amount: amount, payment: { opaqueData: { dataDescriptor:
"COMMON.ACCEPT.INAPP.PAYMENT", dataValue: paymentToken // Secure
token from frontend } }, order: { invoiceNumber: orderId,
description: "Rajarani Restaurant Order" }, customer: { id:
customerId, email: customerEmail } } } };
```

Database Operations (MongoDB)

Transaction & Order Management:

- → Order status updates (pending → paid/failed)
- → Transaction logging with full audit trail
- → Customer payment history
- → Refund tracking capabilities
- → Analytics data collection

transactions collection:

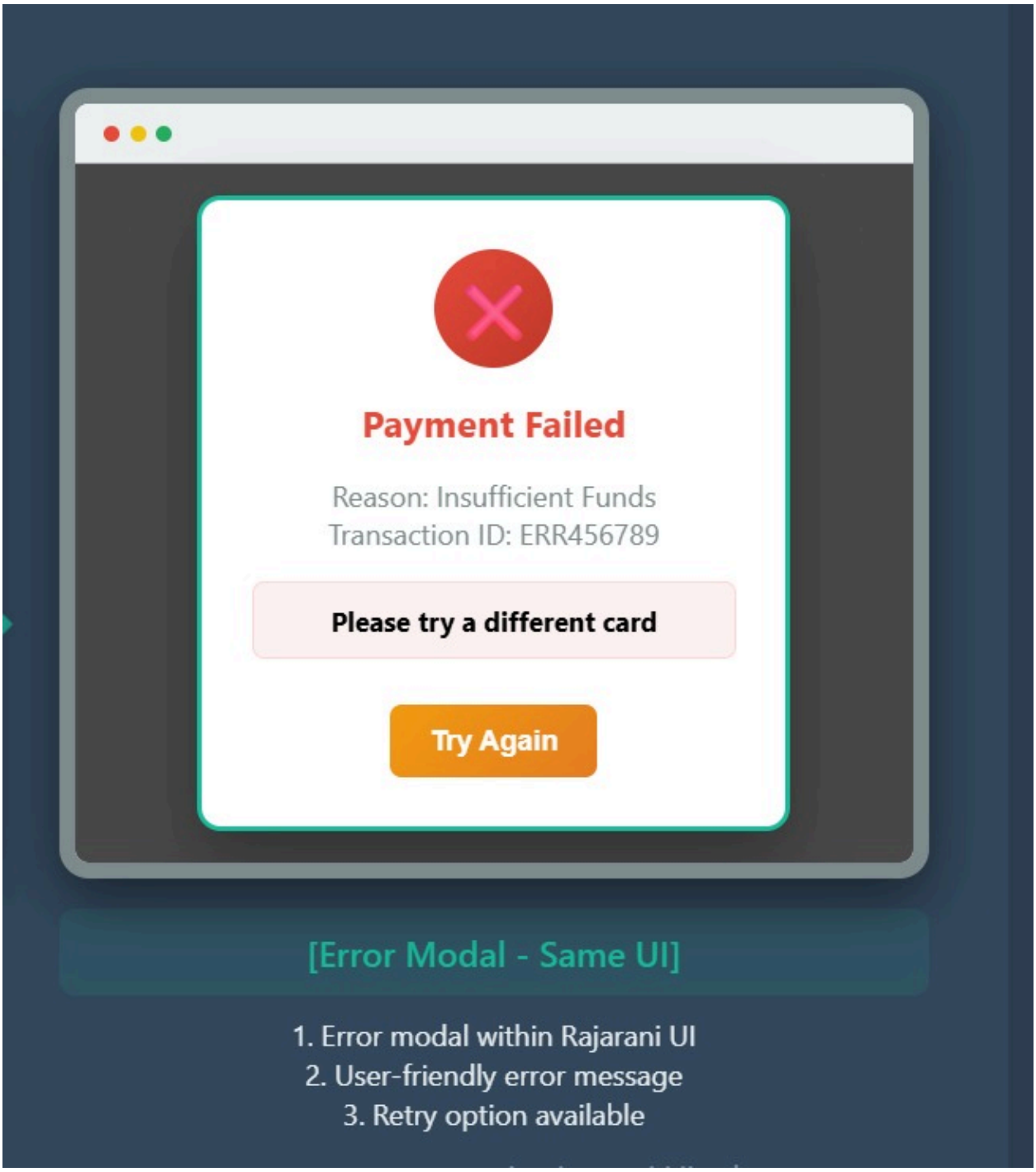
transactionId:	"TXN789123456"
orderId:	ObjectId("...")
amount:	18.01
status:	"approved"
timestamp:	2025-09-10T14:30:00Z
cardLast4:	"3456"

✅ ❌ Success & Failure Response Flows

Payment Successful!

Transaction: TXN789123456

Amount: $18.01


Payment Failed
Reason: Insufficient Funds
Transaction ID: ERR456789
Please try a different card
Try Again
[Error Modal - Same UI]
1. Error modal within Rajarani UI
2. User-friendly error message
3. Retry option available